

# Rate-Quantified BNT Cross-Overlap Decay on the **SectorBNT** Sector Surface

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## 1 Notation

Fix a sector decomposition  $P$  with basis blocks  $\{P.\mathbf{basis}\ j\}_{j=0}^{g-1}$  and a spectral level  $\lambda_j \in \mathbb{C} \setminus \{0\}$  with  $|\lambda_0| > |\lambda_1| > \dots > |\lambda_{g-1}|$  and  $|\lambda_0| = 1$ . The bilinear cross-overlap between basis blocks  $j$  and  $k$  at length  $N$  is

$$O_{j,k}(N) := \mathbf{mpvOverlap}(P.\mathbf{basis}\ j, P.\mathbf{basis}\ k)(N) = \sum_{\sigma} \mathbf{mpv}(P.\mathbf{basis}\ j)(\sigma) \overline{\mathbf{mpv}(P.\mathbf{basis}\ k)(\sigma)}, \quad (1)$$

written `mpvOverlap` in the formalization.<sup>1</sup>

For every ordered pair  $j \neq k$  we are interested in a geometric bound

$$|O_{j,k}(N)| \leq C_{j,k} \eta_{j,k}^N \quad (\forall N \in \mathbb{N}), \quad (2)$$

with  $C_{j,k} \geq 0$  and a rate  $\eta_{j,k} \in [0, 1)$ .

## 2 Cited assertion

The Step 1 argument in the proof of Theorem `thm1` [CPGSV16, line 1182] relies on  $O_{j,k_0}(N) \rightarrow 0$  for  $j \neq k_0$  to discard cross-overlap terms when projecting an assembled BNT identity onto a distinguished basis block  $V^{(N)}(P.\mathbf{basis}\ k_0)$ . The qualitative limit is implicit in the BNT structure: distinct normal blocks satisfy

$$O_{j,k}(N) \xrightarrow{N \rightarrow \infty} 0 \quad (j \neq k). \quad (3)$$

The argument as written does not quantify the rate of decay in (3).

Definition 4.2 of [CPGSV21] packages (3) indirectly under *eventual linear independence* of the basis MPVs and does not state a rate. The formalization records this eventual linear-independence input in the shared sector-decomposition infrastructure,<sup>2</sup> while (3) is provided pointwise by `MPSTensor.mpvOverlap`tendsto`zero`of`not`gaugePhaseEquiv`cast`left`of`irreducible`TP``.<sup>3</sup>

<sup>1</sup>Declaration `MPSTensor.mpvOverlap` in `TNLean/MPS/Overlap/Basic`.

<sup>2</sup>`TNLean/MPS/SharedInfra/SectorDecomposition`.

<sup>3</sup>`TNLean/MPS/Overlap/CastDecay`.

Neither source equips  $\eta_{j,k}$  with a quantitative relation to the spectral weight ratios.

### 3 The retired rate hypothesis

An earlier formal route introduced an explicit rate hypothesis on a sector decomposition  $P$  and a chosen spectral level  $\lambda : \text{Fin } P.\text{basisCount} \rightarrow \mathbb{C}$ . That route has since been retired together with the per-block-projection layer. The mathematical hypothesis it was trying to isolate remains the following: there should exist non-negative families  $C, \eta$  such that for every  $j \neq k$ ,

$$0 \leq C_{j,k}, \quad 0 \leq \eta_{j,k} < 1, \quad \eta_{j,k} \cdot |\lambda_j/\lambda_k| < 1, \quad (4)$$

together with the geometric bound (2) for all  $N$ .

The surviving `SectorBNT` surface does not package such constants. It keeps the raw BNT weights in `SectorDecomposition` and records qualitative consequences in `TNLean/MPS/FundamentalTheorem/SectorBNT/Api`: the coefficient bound `SectorDecomposition.norm`coeff`le`copies`of`norm`weight`le`one`, within-family cross-decay `MPSTensor.IsBNTCanonicalForm.cross`overlap`basis`tendsto`zero`, the combined-family linear-independence criterion `MPSTensor.IsBNTCanonicalForm.combined`family`eventually`li` and the non-vanishing coefficient statement `MPSTensor.IsBNTCanonicalForm.coeff`not`tendsto`zero`at`block`. None of these declarations supplies the constants in (4).

### 4 Where such a hypothesis would be needed

The retired per-block-projection route carried an abstract hypothesis excluding asymptotic cancellation of the assembled overlap against a chosen projection block  $k_0$ . The obstruction analysis in `audits/2026-05-13_cpstv16_ft_path_beta_pr2_5_design.md` shows that for a non-dominant  $k_0$  the discharge requires, after isolating  $V^{(N)}(P.\text{basis } k_0)$  on the right and renormalising by  $\lambda_{k_0}^N$ , that every  $j \neq k_0$  contribute a term whose modulus is bounded by

$$|O_{j,k_0}(N)| \cdot |\lambda_j/\lambda_{k_0}|^N \leq C_{j,k_0}(\eta_{j,k_0} \cdot |\lambda_j/\lambda_{k_0}|)^N. \quad (5)$$

The right-hand side of (5) converges to 0 exactly when (4) holds. The qualitative limit (3) alone gives  $\eta_{j,k_0} < 1$  but not the comparison  $\eta_{j,k_0} \cdot |\lambda_j/\lambda_{k_0}| < 1$ , which is the content of the new hypothesis.

### 5 Classification

*Open gap.* The rate condition (4) is a structural input beyond what [CPGSV16, §II] and [CPGSV21, Definition 4.2] make explicit. Its rate is in principle derivable from the spectral gap of the mixed transfer operator that powers `mpvOverlap`endsto`zero`of`not`gaugePhaseEquiv`cast`left`of`irreducible`TP`, but the compatibility inequality  $\eta_{j,k} \cdot |\lambda_j/\lambda_k| < 1$  is not implied by the BNT structure and must be either imposed or established analytically.

The analytic derivation from the transfer matrix spectral gap is the subject of *Option 2* in the design memo `audits/2026-05-13_cpsv16_ft_path_beta_pr2_5_design.md` and is not addressed in this note. The current `SectorBNT` route instead keeps the coefficient comparison and the combined-family linear independence in `TNLean/MPS/FundamentalTheorem/SectorBNT`; any future rate-quantified argument should be stated against that surviving surface, not against the retired per-block-projection layer.

## References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.